Summarization and graphical presentation of data(1)

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How to present data

- Frequency distribution
- Graphical representation

Frequency distribution

- The number of times a particular value or range of values appears in the dataset

- Arranging data into homogeneous/similar group

Frequency distribution

- **For example, the brand of phone of randomly selected 10 students,**

  Samsung, Apple, Apple, One-plus, Huawei, Samsung, Apple, Apple, Huawei, One-plus.

Brand	Frequency
Samsung	2
Apple	4
One-plus	2
Huawei	2

Class work

- Blood group from 10 students,

  O, A, B, O, AB, B, A, O, A, AB

.

Blood group	Frequency
O	
A	
B	
AB	

Types of FD

- Frequency distribution of Quantitative data
- Frequency distribution of Qualitative data

FD for Quantitative

- There are five steps in constructing a frequency distribution table for quantitative data

1. Choose the number of classes ($k = \sqrt{n}$)
2. Class interval ($i = \frac{Highest\ value - Lowest\ value}{k}$)
3. Set the individual class/class limits
4. Tally
5. Frequency

FD for Quantitative

- Distance of University from home (KM) for randomly selected students,

$$3, 8, 12, 9, 14, 6, 10, 11, 7, 5, 7, 16, 8, 13$$

Here, the number of classes, $k = \sqrt{n} = \sqrt{14} = 3.74 \sim 4$

Class interval is, $i = \frac{H-L}{K} = \frac{16-3}{4} = 3.25 \sim 4$

Class	Tally	Frequency
2-6	||	2
6-10	𝍸 |	6
10-14	|||	4
14-18	|||	2

FD for Quantitative

- Relative frequency
- $Proportion = f_i/N$

Class	Tally	Frequency	Relative frequency
2-6	||	2	2/14=0.14
6-10	𝍸 |	6	0.43
10-14	||	4	0.29
14-18	|||	2	0.14
N=14			

FD for Quantitative

- Percentage frequency
- Percentage = f_i/N*100

Class	Tally	Frequency	Relative frequency	Percentage frequency
2-6	||	2	2/14=0.14	0.14×100=14%
6-10	𝍸 |	6	0.43	43%
10-14	||	4	0.29	29%
14-18	|||	2	0.14	14%

FD for Quantitative

- Cumulative frequency
- Cumulative frequency, F=$\sum f_i$

Class	Tally	Frequency	Relative frequency	Percentage frequency	Cumulative frequency
2-6	||	2	0.14	14%	2
6-10	𝍸 |	6	0.43	43%	2+6=8
10-14	||	4	0.29	29%	12
14-18	|||	2	0.14	14%	14

FD for Quantitative

- Relative cumulative frequency
- Relative cumulative frequency= F_i/N

Class	Tally	Frequency	Relative frequency	Percentage frequency	Cumulative frequency	Relative cumulative frequency
2-6	||	2	0.14	14%	2	0.14
6-10	𝍸 |	6	0.43	43%	8	0.57
10-14	||	4	0.29	29%	12	0.86
14-18	|||	2	0.14	14%	14	1

Class work

- Age (in years) of 30 randomly selected men,

43,66,54,26,64,36,16,31,36,21,53,27,58,29,25,50,24,14,43,
34,32,26,15,22,49,55,26,11,57,15.

- Make frequency distribution with appropriate class number and class interval. Also, estimate class midpoint.

Class work

Age (in Years)	Midpoint	Frequency	Relative frequency	Cumulative frequency	Relative Cumulative freq.
10-20	10+20/2=15	5	0.167	5	0.167
20-30	25	9	0.3	14	
30-40	35	5			
40-50		3			
50-60		6			
60-70		2			
		N=30			

Class work

If the,

- **Class number**
- **or class interval**
- **or both**

defined earlier then we have to follow it.

FD for Qualitative

- **For example, the brand of phone of randomly selected 10 students,**

  Samsung, Apple, Apple, One-plus, Huawei, Samsung, Apple, Apple, Huawei, One-plus.

Brand	Tally	Frequency
Samsung	||	2
Apple	||||	4
One-plus	||	2
Huawei	||	2
Total		10

Thank You